The Clebsch cubic program — IV

A binary $[78, 36, 12]$ code from the passant lines of a conic over $\mathbb{F}_{13}$

Tavis Rudd

August 2026

Abstract

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2, 13)$, and form the binary incidence code on its 78 internal points using the 78 passant lines. We prove that this code has parameters $[78, 36, 12]_2$. It has exactly 364 minimum words, in four $\mathrm{PGL}(2, 13)$-orbits of size 91, and each orbit spans the code. More strongly, the minimum supports reconstruct the passant incidence matrix and the six-class elliptic association scheme; their coordinate-permutation automorphism group is exactly $\mathrm{PGL}(2, 13)$. The proof combines a tangent-product exclusion in weight eight, two exhaustive but compact syndrome-disjointness checks in weight ten, and structural arguments in the binary association algebra for spanning and reconstruction.

Keywords. binary incidence code; finite conic; passant line; minimum word; association scheme; reconstruction.

1 Introduction and statement

Incidence codes often retain less geometry than the incidence structures that define them. Row operations erase the choice of parity checks, and a minimum-distance computation need not reveal why the first codewords occur. Here the minimum words recover both the parity checks and the association scheme on the coordinates.

The code belongs to the fixed-conic LDPC family introduced by Droms–Mellinger–Meyer [2]. Their general bounds, as recorded in the later comparison table of Ma–Liu–Tian [5, Table 1], specialize here to $8 \le d \le 12$. Madison and Wu proved the general nullity formula $(q-1)^2/4$ [4, Corollary 6.3]. Hollmann and Xiang constructed the elliptic association scheme afforded by the same conic action [3, Sections 3–5]. We determine the upper endpoint of the distance interval at $q = 13$, classify its minimum words, and recover the incidence and relation structures from their unlabeled support hypergraph.

Fix a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2, 13)$. A line is *passant* when it is disjoint from $\mathcal{C}(\mathbb{F}_{13})$. Conic polarity identifies the 78 internal points with the 78 passant lines. Let M be their binary incidence matrix and put

$$K = \ker_{\mathbb{F}_2} M \subseteq \mathbb{F}_2^{78}.$$

The coordinate set is the set of internal points. Automorphisms below are coordinate permutations preserving the indicated code or minimum-support system.

Let

$$\mathcal{H} = \{\mathrm{supp}(c) : c \in K,\ \mathrm{wt}(c) = 12\}$$

be the minimum-support hypergraph on the unlabeled coordinate set of K.

Theorem 1.1 (Minimum-layer reconstruction). *The code K has parameters $[78, 36, 12]_2$ and exactly 364 minimum words. Under $\mathrm{PGL}(2, 13)$, those words form four orbits of size 91: one has stabilizer isomorphic to S_4, and three have dihedral stabilizer of order 24. Every one of the four orbits spans K.*

From the abstract hypergraph $\mathcal{H}$, pair and triple concurrence reconstruct the six relations of the elliptic association scheme on the coordinate set. Among the passant seven-cliques, the 78 with zero triple concurrence are exactly the supports of the rows of M. Consequently $\mathcal{H}$ reconstructs M up to independent coordinate and row permutation, and

$$\mathrm{Aut}(K) = \mathrm{Aut}(\mathcal{H}) = \mathrm{PGL}(2, 13)$$

in its symmetric-square action on the 78 internal points.

Reconstruction here is a permutation statement. It does not choose an equation for the conic, label the projective points, or order the parity checks. The proof has three finite leaves: a five-row clique closure in weight eight, two syndrome-disjointness checks in weight ten, and an exact orbit/concurrence replay for the minimum layer. The reductions to those leaves and the association-algebra argument are mathematical proofs in the text.

2 Distance twelve

We use the model $\mathcal{C} : XZ - Y^2 = 0$. Every row and column of M has weight seven, and conic polarity identifies the two index sets. The Madison–Wu formula gives $\dim K = 36$; direct binary elimination is an independent check of rank 42.

Summing all row equations gives the all-one functional, so every word of K has even weight. If a nonzero word contains an internal point P, each of the seven passants through P contains a second support point. Those seven points are distinct. Thus every nonzero word has weight at least eight.

2.1 The tangent obstruction in weight eight

Suppose that a word has weight eight. The preceding parity argument forces its support to be an eight-arc: every join of two support points is passant, and every passant pencil at a support point is saturated. The seven remaining lines through each support point are therefore both the combinatorial tangents to the arc and the conic secants through that point.

For an internal point P, put

$$T_P(X) = \prod_{\substack{\ell \ni P \\ \ell \text{ secant to } \mathcal{C}}} \ell(X).$$

For a pairwise-passant triple define

$$h(P, Q, R) = \frac{T_P(Q)T_Q(R)T_R(P)}{T_P(R)T_R(Q)T_Q(P)}.$$

Segre's lemma of tangents in the coordinate-free form of Ball–Lavrauw [1, Lemma 27] gives $h(P, Q, R) = 1$ for every triple in the hypothetical support. Fix $P = (1 : 0 : 2)$. Its other seven points would form a clique in the graph on the 42 passant-join neighbors of P, where Q and R are adjacent when QR is passant and $h(P, Q, R) = 1$.

An order-14 element of the stabilizer of P splits the vertices into three cyclic orbits A_i, B_i, C_i, indexed by $\mathbf{Z}/14$. Direct substitution in the product above gives

orbit pair	$j - i \pmod{14}$
AA	$4, 6, 8, 10$
AB	$6, 7, 11, 12$
AC	$1, 3$
BB	$6, 8$
BC	$3, 5, 6, 8, 9, 11$
CC	$2, 4, 10, 12.$

Up to simultaneous translation, every four-clique is represented by one of the following five rows; the right column gives its unique extension:

four-clique	unique extension
A_0, B_6, B_{12}, C_1	C_3
A_0, B_6, B_{12}, C_3	C_1
A_0, B_6, C_1, C_3	B_{12}
A_0, B_{12}, C_1, C_3	B_6
B_0, B_6, C_9, C_{11}	$A_8.$

They close to fourteen translates of one five-clique, and none extends. The local clique number is five, so a seven-clique cannot occur. This excludes weight eight.

2.2 The two weight-ten profiles

Fix one point in a hypothetical weight-ten support. Let s of the other nine points be joined to it by secants. The remaining $9 - s$ points occur in positive odd numbers in its seven passant fibres. Hence s is even and $9 - s \geq 7$, so $s = 0$ or $s = 2$. The profiles are

$$(3, 1, 1, 1, 1, 1, 1; 0), \qquad (1, 1, 1, 1, 1, 1, 1; 2),$$

where the last entry is the number of secant neighbors.

Write m_Q for the 78-bit incidence column at Q. For the first profile and each choice of exceptional fibre, the finite check forms two canonically sorted partial-XOR sets of sizes 216 and 4320. For the second profile the sizes are 216 and 771024. In every case the two sets are disjoint. The raw domains have sizes

$$7\binom{6}{3}6^6 = 6{,}531{,}840, \qquad 6^7\binom{35}{2} = 166{,}561{,}920.$$

Thus the split covers every support in each profile and proves that no candidate has zero syndrome. An independent dynamic program reconstructs the conic and incidence columns, builds the full-fibre XOR sets with a different partition, and obtains the same two empty intersections.

Finally, the twelve internal points

$$(1:0:2), (1:3:2), (1:4:5), (1:1:8), (1:4:8), (1:1:7),$$
$$(1:7:12), (1:3:3), (1:9:11), (1:10:11), (1:0:5), (1:8:7)$$

have zero syndrome. Their stabilizer is dihedral of order 24. Therefore $d(K) = 12$.

3 Minimum words and the elliptic scheme

The exact replay first fixes one support point. Pencil parity leaves $s = 0, 2, 4$ secant neighbors, and a meet-in-the-middle enumeration of all three profile families finds exactly 56 weight-twelve words through the fixed point. The four proposed projective orbits have disjoint fixed-point slices of sizes 14, and those slices equal the 56-word exhaustive list. The order-28 stabilizer of the fixed point acts transitively on each slice; the stabilizer of a support inside it has order 2. Transitivity, or equivalently the count $56 \cdot 78/12 = 364$, proves global exhaustion.

The same replay generates each orbit from a projective representative and checks every support against M. The stabilizers have order 24, so each orbit has size $2184/24 = 91$. Their element-order profiles identify one stabilizer with S_4 and the other three with dihedral groups of order 24.

For a pair or triple of coordinates, its concurrence is the number of the 364 minimum supports containing it. Pair concurrences are $7, 9, 12$ when the joining line is passant and $6, 8$ when it is secant. The full triple- concurrence histograms are distinct on the six $\mathrm{PGL}(2, 13)$-orbitals. Hence these intrinsic counts recover all six colors of the elliptic association scheme.

After recovering the passant color, form its seven-cliques. Exactly 78 have zero triple concurrence, and direct incidence comparison identifies them with the supports of the 78 rows of M. Both tests use only the minimum-support hypergraph after its coordinate set is supplied. They therefore define a reconstruction map independent of the original projective labels.

4 Spanning and automorphisms

For internal points $P = (x_P : y_P : z_P)$ and Q, set

$$\Delta(P) = y_P^2 - x_P z_P, \qquad \beta(P, Q) = 2y_P y_Q - x_P z_Q - z_P x_Q,$$

and

$$\rho(P, Q) = \frac{\beta(P, Q)^2}{\Delta(P)\Delta(Q)}.$$

On distinct internal points, ρ takes the six values $0, 1, 3, 9, 10, 12$. Its level relations are the elliptic scheme; write A_r for their adjacency matrices. Polarity identifies A_0 with M, up to row order.

The needed part of the integral intersection table reduces modulo two to

$$\begin{aligned} A_0^2 &= I + A_9 + A_{10} + A_{12}, \\ A_0 A_r &= 0 \quad (r = 9, 10, 12), \\ A_9^2 &= A_{10}, \qquad A_{10}^2 = A_{12}, \qquad A_{12}^2 = A_9. \end{aligned} \tag{1}$$

These are parity reductions of intersection numbers, checked by fixing one point and substituting one representative of each relation.

Put $B = A_9$. Since $A_0 B = 0$, the image of B lies in K. If $x \in K$ and $Bx = 0$, then (1) gives

$$0 = A_0^2 x = (I + B + B^2 + B^4)x = x.$$

Thus B is injective on the 36-dimensional space K, so $\operatorname{im} B = K$, and B, B^2, B^4 all have rank 36. If N_i is the 91-by-78 support matrix of the i-th minimum-word orbit, direct pair counting in one representative gives

$$(N_1^\top N_1, N_2^\top N_2, N_3^\top N_3, N_4^\top N_4) = (A_9, A_9, A_{12}, A_{10}).$$

Every N_i therefore has rank at least 36. Its rows lie in K, so each orbit spans K.

It remains to determine the scheme automorphisms. Use the four anchors

$$P_0 = (1:0:2), \quad P_1 = (1:1:7), \quad P_2 = (1:0:7), \quad P_3 = (1:1:3).$$

The first three have pair-relation pattern $(10,3,9)$. There are $78 \cdot 14 \cdot 2 = 2184$ ordered triples with this pattern: A_{10} has valency 14 and $p_{3,9}^{10} = 2$. Substitution in the symmetric-square action shows that the stabilizer of (P_0, P_1, P_2) in $\mathrm{PGL}(2,13)$ is trivial. Since the group also has order 2184, it acts simply transitively on these triples.

For a fixed triple, P_3 is the unique point with relation signature $(3,1,9)$. The four values

$$\big(\rho(P,P_0), \rho(P,P_1), \rho(P,P_2), \rho(P,P_3)\big),$$

with equality to an anchor recorded on the diagonal, distinguish all 78 points. Consequently a scheme automorphism can be composed with a unique element of $\mathrm{PGL}(2,13)$ to fix the first three anchors; it then fixes the fourth and every point. Thus the reconstructed scheme has automorphism group $\mathrm{PGL}(2,13)$.

An automorphism of K permutes its minimum supports. Conversely an automorphism of $\mathcal{H}$ preserves K, because any one of the four minimum orbits spans it. The two coordinate-permutation groups therefore coincide. This completes the proof of Theorem 1.1.

5 Human proof and exact finite evidence

The proof above is a human structural proof: parity and pencil geometry make the low-weight reductions, the tangent product controls weight eight, the association algebra explains spanning, and concurrence and anchors explain reconstruction and rigidity. Computation is not the proof architecture. It records discovery and checks only finite leaves that have not admitted useful conceptual compression.

The public evidence directory contains four paper-owned objects. A profile generator checks the canonical weight-ten finite data. An independent dynamic program reconstructs the incidence matrix and repeats both exclusions. The main exact replay checks the clique closure, rank, minimum layer, concurrence reconstruction, spanning identities, and automorphism anchors. The aggregate entry point `verify_evidence.py` first checks all recorded byte counts and SHA-256 hashes and then runs the three replays. Its adjacent README gives the individual filenames and commands.

The trusted boundary is explicit. The shared Lean library mirrors the normalized conic, the incidence code, parity-profile reductions, association-algebra deductions, tangent graph, reconstruction semantics, and four-anchor closure used by the human argument. A paper-owned Lean package checks the remaining finite bulk: fourteen weight-ten syndrome shards, exact bit-row rank 42, four projective kernel orbits of size 91 and span rank 36, the binary association identities, pair-color recovery, the zero-triple signatures of the 78 geometric rows, uniqueness of the recovered row family, and the fixed-point weight-twelve exhaustion. The last check implements the four pencil-profile domains by a syndrome-indexed meet in the middle and identifies their 56 solutions with the four disjoint 14-support orbit slices. Explicit recovery and expansion maps transport the 42-column elimination certificate to the semantic incidence map, so the formal aggregate also proves dimension 36. A generic Boolean-parity bridge, four one-product relation shards, and four orbit shards transport the concrete association identities and Gram matrices into ordinary matrices over $\mathbf{F}_2$. The abstract association-kernel argument then proves that the row space of every displayed orbit is exactly $\ker A_0$; polarity is the row permutation identifying this kernel with the code. The terminal declaration is `PassantCodeQ13.Gates.Main.minimumOrbitsSpanRhoZeroKernel`.

These checks use native evaluation and expose the corresponding declaration-local axioms in the pinned toolchain.

Row-family uniqueness is compressed to local extension pools. For each three-point seed, retain the points that extend it to an admissible four-set. Every non-row seed has at most ten such extensions. Seven first-index residue leaves check every four-subset of these pools, while a symbolic coverage argument shows that every admissible seven-set occurs in one of those checks. The three-point shortcut itself is false: there are (1456) non-row passant triangles with zero triple concurrence, so the four-extension test is essential rather than a computational convenience. The terminal declaration is `PassantCodeQ13.Gates.Main.recoveredRowFamilyIsUnique`. The independent replay follows a different route by enumerating all 1716 passant seven-cliques before selecting the 78 zero-concurrence rows.

For weight eight, the shared formalization derives saturation of the seven passant pencils from the semantic row equations, identifies the 42 passant-join neighbors of $(1 : 0 : 2)$ with the three cyclic graph orbits, and checks that graph adjacency is exactly passant join together with tangent holonomy one. The five-row unique-extension certificate then proves that the seven companions cannot exist. This transport takes Segre's lemma of tangents and the preceding arc identification as the classical geometric input stated in the proof. Its terminal declaration is `RelativeConicArcs.Gates.PassantCodeQ13.weightEight_semantic_transport`.

For weight ten, the shared formalization now starts from an arbitrary codeword and any point in its support. The seven actual passant rows through that point have odd support fibres; the other support points are exactly its secant neighbors. Partitioning the remaining nine points then gives either one triple fibre and six singleton fibres with no secant neighbor, or seven singleton fibres with two secant neighbors. Its terminal declaration is

```
RelativeConicArcs.Gates.PassantCodeQ13.
arbitrary_weightTen_profile_transport.
```

These checks support the finite leaves; they do not replace the structural proof. Projective transitivity and the double count transport the checked fixed-point exhaustion to the full weight-twelve layer in the human proof. The formal package does not yet prove uniqueness of the reconstructed row family or the four-anchor classification of every coordinate automorphism. The complete theorem is therefore presented first and foremost as a human proof, with exact finite evidence at its uncompressed leaves; its formal subclaims have the narrower scope just stated.

6 Conclusion

The minimum layer determines more than the parameters of the code: it recovers the finite geometry that defined the parity checks. The result therefore gives a concrete reconstruction theorem for a binary incidence code, with the irreducibly finite exclusions isolated from the association- algebra mechanism that explains spanning and symmetry.

References

[1] S. Ball and M. Lavrauw, Arcs in finite projective spaces, *EMS Surveys in Mathematical Sciences* **6** (2019), 133–172. arXiv:1908.10772.

[2] S. V. Droms, K. E. Mellinger, and C. Meyer, LDPC codes generated by conics in the classical projective plane, *Designs, Codes and Cryptography* **40** (2006), 343–356. doi:10.1007/s10623-006-0022-6.

[3] H. D. L. Hollmann and Q. Xiang, Association schemes from the action of $\mathrm{PGL}(2, q)$ fixing a nonsingular conic in $\mathrm{PG}(2, q)$, *Journal of Algebraic Combinatorics* **24** (2006), 157–193. doi:10.1007/s10801-006-0005-8; arXiv:math/0503573.

[4] A. L. Madison and J. Wu, On binary codes from conics in $\mathrm{PG}(2, q)$, *European Journal of Combinatorics* **33** (2012), 33–48. doi:10.1016/j.ejc.2011.08.001; arXiv:1104.0324.

[5] L. Ma, S. Liu, and Z. Tian, The binary codes generated from quadrics in projective spaces, *AIMS Mathematics* **9** (2024), 29333–29345. doi:10.3934/math.20241421.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.